

ROBE, Australia

WMO: 958160

Lat: 37.1775S

Lon: 139.8053E

Elev: 11

StdP: 14.69

Time Zone: 9.50 (AUA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 35.6	(c) 37.9	(d) 32.8	(e) 27.3	(f) 43.2	(g) 34.7	(h) 29.5	(i) 44.0	(j) 28.3	(k) 55.6	(l) 25.8	(m) 55.3	(n) 2.4	(o) 50	(p) 0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 18.9	(c) 90.3	(d) 65.7	(e) 85.5	(f) 65.0	(g) 80.9	(h) 63.8	(i) 68.9	(j) 81.4	(k) 67.1	(l) 79.4	(m) 65.6	(n) 76.5	(o) 13.3	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	92.7	71.4	63.3	86.9	70.2	61.7	82.3	68.8	33.1	82.2	31.6	79.8	30.4	76.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	22.5	20.5	31.0	98.5	2.0	3.8	29.5	101.2	28.3	103.5	27.2	105.6	25.7	108.4
			30.6	72.2	2.0	2.3	29.1	73.9	27.9	75.2	26.8	76.5	25.3	78.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.6	65.0	64.6	62.5	58.7	55.3	51.1	51.0	52.0	53.5	56.2	59.6	62.4
	DBStd	7.03	6.19	5.42	6.08	4.86	4.30	3.94	3.55	3.56	4.07	4.97	5.91	5.78
	HDD50	107	0	0	0	0	7	33	33	20	11	3	0	0
	HDD65	2957	74	66	123	199	304	416	435	404	344	280	187	125
	CDD50	2885	463	409	388	261	170	67	63	81	116	194	290	383
	CDD65	262	72	55	47	10	2	0	0	0	0	6	27	43
	CDH74	2882	823	516	450	112	8	0	0	1	4	112	322	534
	CDH80	1138	387	202	174	22	1	0	0	0	0	28	106	218
(14) Wind	WSAvg	10.1	11.1	10.7	9.7	8.5	9.1	8.5	10.5	10.6	10.3	10.4	10.7	10.6
(15) Precipitation	PrecAvg	25.2	0.7	0.7	1.2	1.8	2.7	3.7	4.2	3.6	2.4	1.7	1.3	1.1
	PrecMax	33.5	5.2	3.7	4.9	4.5	5.2	7.0	7.6	7.6	5.2	4.2	3.1	5.3
	PrecMin	14.2	0.0	0.0	0.0	0.3	0.4	0.6	1.6	1.3	0.8	0.3	0.2	0.1
	PrecStd	4.4	0.9	0.7	1.0	1.2	1.3	1.3	1.5	1.5	1.0	0.9	0.8	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.4	94.2	92.1	83.4	72.8	63.9	62.3	69.1	72.9	83.9	89.0	92.5
		MCWB	67.2	68.1	64.9	62.6	57.7	57.6	55.9	55.8	58.4	62.5	65.1	65.3
	2%	DB	90.8	86.2	84.9	77.1	67.5	61.1	59.7	63.1	68.3	76.4	82.8	86.5
		MCWB	66.1	65.6	64.6	60.1	57.6	55.8	54.1	54.5	55.5	59.5	64.2	64.4
	5%	DB	83.2	80.4	78.6	71.8	64.4	59.6	58.2	60.2	64.0	71.2	76.8	80.3
		MCWB	65.4	64.8	63.3	59.4	58.2	54.9	53.2	54.0	55.4	58.4	63.0	63.9
	10%	DB	76.6	74.8	72.9	67.8	62.3	58.2	57.0	58.2	61.1	66.0	70.8	73.8
		MCWB	64.1	63.6	62.2	59.0	57.1	53.8	52.5	52.9	54.7	57.3	60.7	62.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	69.8	70.2	64.8	62.7	59.8	57.8	58.4	60.4	64.4	68.4	69.7
		MCDB	88.9	83.6	80.4	74.1	65.6	61.0	60.6	62.7	66.3	75.9	81.9	79.6
	2%	WB	68.9	67.8	67.1	63.2	60.6	57.4	55.6	56.2	58.3	61.7	65.8	66.9
		MCDB	81.5	81.6	77.3	70.7	63.9	59.5	58.1	61.1	64.2	72.0	79.4	80.5
	5%	WB	66.8	66.1	65.0	61.7	59.2	55.8	54.3	54.7	56.7	59.5	63.5	64.7
		MCDB	78.3	77.3	74.2	68.1	63.0	58.4	57.1	59.1	62.1	68.4	73.7	76.1
	10%	WB	64.7	64.3	63.3	60.2	57.8	54.5	53.2	53.5	55.3	57.9	61.2	62.9
		MCDB	75.9	73.1	71.5	66.2	61.8	57.3	56.2	57.6	60.6	65.0	70.1	72.7
(35) Mean Daily Temperature Range	MDBR		18.9	17.8	18.6	17.9	13.7	13.8	11.6	12.4	14.4	17.1	17.4	18.0
	5% DB	MCDBR	31.3	27.9	28.4	26.3	17.3	15.4	12.5	16.3	20.9	28.3	28.5	30.8
		MCWBR	12.2	11.5	11.8	12.9	11.4	11.6	9.2	10.8	12.3	14.7	13.1	12.8
	5% WB	MCDBR	24.3	22.7	21.9	18.9	14.0	13.4	10.7	14.2	17.3	23.5	25.1	24.1
		MCWBR	11.8	10.7	11.2	11.6	10.7	11.2	8.9	10.3	11.6	13.6	12.7	11.8
(40) Clear-Sky Solar Irradiance	taub		0.340	0.338	0.326	0.318	0.308	0.304	0.309	0.319	0.340	0.358	0.337	0.337
	taud		2.536	2.555	2.587	2.596	2.604	2.599	2.580	2.543	2.474	2.418	2.515	2.533
	Ebn at Noon		315	309	300	284	267	259	264	279	290	298	314	318
	Edh at Noon		35	33	29	25	22	21	22	26	32	37	35	35
(44) All-Sky Solar Radiation	RadAvg		2373	2050	1625	1129	753	613	662	951	1345	1782	2121	2342
	RadStd		115	100	68	72	41	38	36	49	110	118	88	131

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-205	+245	+65

Nomenclature: See separate page